

Safety data sheet

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BASF Safety data sheet
Date / Revised: 03.08.2023
Product: **Lutensol® TO 5**

Version: 8.0

(30044057/SDS_GEN_TH/EN)

Date of print: 31.05.2026

1. Substance/preparation and manufacturer/supplier identification

Product name:
Lutensol® TO 5

Use: Raw material for the chemical-technical industry

Manufacturer/supplier:

BASF (Thai) Limited
23rd Floor, Emporium Tower, 622, Sukhumvit 24 Rd.,
Klongton, Klongtoey, Bangkok 10110, THAILAND
Telephone: +66 2624-1999
Telefax number: +66 2664-9254
E-mail address: Thailand-SDS-info@basf.com

Emergency information:

International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

Skin corrosion/irritation: Cat.3

Serious eye damage/eye irritation: Cat.2B

Hazardous to the aquatic environment - acute: Cat.2

Hazardous to the aquatic environment - chronic: Cat.3

Label elements and precautionary statement:

Signal Word:

Warning

Hazard Statement:

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H320	Causes eye irritation.
H316	Causes mild skin irritation.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

Precautionary Statements (Prevention):

P273	Avoid release to the environment.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337 + P313	If eye irritation persists: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Other hazards which do not result in classification:

No specific dangers known, if the regulations/notes for storage and handling are considered.

This classification is based on the current CESIO recommendations. This surfactant complies with the biodegradability criteria as laid down in Regulation (EC) No.648/2004 on detergents. Data to support this assertion are held at the disposal of the competent authorities of the Member States and will be made available to them at their direct request or at the request of a detergent manufacturer.

3. Composition/information on ingredients

Chemical nature

Substance nature: Substance

Poly(oxy-1,2-ethanediyl), .alpha.-tridecyl-.omega.-hydroxy-, branched
CAS Number: 69011-36-5

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

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On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:

harmful vapours, carbon oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For large amounts: Dike spillage. Pump off product.

For residues: Pick up with suitable absorbent material.

Dispose of absorbed material in accordance with regulations.

Additional information: High risk of slipping due to leakage/spillage of product. Forms slippery surfaces with water.

7. Handling and Storage

Handling

Protect against moisture. Shut containers immediately after taking product because product takes up the humidity of air. No special measures necessary provided product is used correctly.

Protection against fire and explosion:
No special precautions necessary.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4301 (V2), Stainless steel 1.4306 (V2A), Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4439, Stainless steel 1.4539, Stainless steel 1.4541, Stainless steel 1.4571, Stove-lacquer RDL 50

Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

Storage stability:

Storage temperature: $\leq 70\text{ }^{\circ}\text{C}$

The packed product is not damaged by low temperatures or by frost. Bulk must be protected from solidification.

Protect from temperatures above: $70\text{ }^{\circ}\text{C}$

Properties of the product change irreversibly on exceeding the limit temperature.

8. Exposure controls and personal protection

Components with occupational exposure limits

No substance specific occupational exposure limits known.

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for higher concentrations or long-term effect: (Combination filter EN 14387 A-P2)

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

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Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless	
Odour:	product specific	
Odour threshold:	not determined	
pH value:	approx. 7 (50 g/l, 20 °C)	(DIN EN 1262)
Freezing point:	< 5 °C	(DIN ISO 2207)
Boiling point:	> 250 °C	(estimated)
Flash point:	approx. 175 °C	(ISO 2592)
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability (solid/gas):	not self-igniting	
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	> 300 °C	(other)
Thermal decomposition:	> 350 °C	(DTA)
Self ignition:	not self-igniting	
Self heating ability:	It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Vapour pressure:	< 0.1 hPa (20 °C)	(estimated)

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Density:	approx. 0.96 g/cm ³ (20 °C)	(DIN 51757)
	approx. 0.93 g/cm ³ (70 °C)	(DIN 51757)
Relative density:	No data available.	
Relative vapour density (air):	not determined	
Solubility in water:	insoluble	
Miscibility with water:	partly miscible	
Hygroscopy:	hygroscopic	
Solubility (qualitative) solvent(s):	Ethanol soluble	
Partitioning coefficient n-octanol/water (log Pow):	not applicable	
Surface tension:	approx. 27 mN/m (23 °C; 1 g/l)	(other)
Viscosity, dynamic:	< 20 mPa.s (60 °C)	(DIN EN 12092)
	approx. 80 mPa.s (23 °C)	(DIN EN 12092)

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

See SDS section 7 - Handling and storage.

Thermal decomposition: > 350 °C (DTA)

Substances to avoid:

caustics, halogens, Alkalines, acids, reactive chemicals

Corrosion to metals: Corrosive effects to metal are not anticipated.

Hazardous reactions:

No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

Reactivity:

No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): > 5,000 mg/kg (BASF-Test)

Literature data.

Acute dermal toxicity

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Literature data.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

Irritation

Experimental/calculated data:

Skin corrosion/irritation rabbit: Slightly irritating. (OECD Guideline 404)

Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

Literature data.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on the structure, there is no suspicion of a skin-sensitizing potential.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on the structure, there is no suspicion of a mutagenic effect.

Carcinogenicity

Assessment of carcinogenicity:

Based on the structure there is no suspicion of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Developmental toxicity

Assessment of teratogenicity:

Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

The information available on the product provides no indication of toxicity on target organs after repeated exposure.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Ecotoxicity

Toxicity to fish:

LC50 (96 h) > 1 - 10 mg/l, *Leuciscus idus* (DIN 38412 Part 15)

Literature data.

Aquatic invertebrates:

EC50 (48 h) > 1 - 10 mg/l, *Daphnia magna* (DIN 38412 Part 11)

Literature data.

Aquatic plants:

EC50 (72 h) > 1 - 10 mg/l (growth rate), *Scenedesmus subspicatus* (DIN 38412 Part 9)

acute Effect Literature data.

EC10 (72 h) > 0.1 - 1 mg/l (growth rate), *Skeletonema costatum* (ISO/DIS 10253)

long-term effect The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Literature data.

Microorganisms/Effect on activated sludge:

EC10 (17 h) > 2,500 mg/l (DIN 38412 Part 8)

Chronic toxicity to fish:

No observed effect concentration > 1 mg/l, Fish

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Literature data.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration, > 1 mg/l, *Daphnia magna*

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Literature data.

Assessment of terrestrial toxicity:

No data available concerning terrestrial toxicity.

Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is possible.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):
Readily biodegradable (according to OECD criteria).

Elimination information:
> 60 % CO₂ formation relative to the theoretical value (28 d) (OECD 301B; ISO 9439; 92/69/EEC, C.4-C) Readily biodegradable (according to OECD criteria).

Sum parameter

Chemical oxygen demand (COD): 2,438 mg/g

Bioaccumulation potential

Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations. Do not release untreated into natural waters.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Untampered packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

UN number or ID number	Not classified as a dangerous good under transport regulations
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for	None known

user

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

15. Regulatory Information

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

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